

Isobutanol Synthesis from Syngas over Zn-Cr Catalyst: Effect of Zn/Cr Element Ratio

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This report demonstrated the effect of Zn/Cr element ratio on Zn–Cr catalyst for isobutanol formation from the reaction of carbon monoxide hydrogenation. The catalysts of different Zn/Cr element ratios including 1:3, 1:1, 3:1 and 5:1 have been prepared by a facile coprecipitation approach and their activity evaluation of isobutanol formation by carbon monoxide hydrogenation were tested. The Zn₁Cr₁ catalyst of Zn/Cr element ratio being 1:1 presented outstanding activity for iso-butanol formation from the carbon monoxide hydrogenation, carbon monoxide conversion and isobutanol selectivity reached to the maximum to ~32 % and ~24 %, respectively.

Multi-Characterization including TEM, XRD, BET, XAS, H₂-TPR, CO₂-TPD and XPS were proceed to investigate the physico-chemical property of catalysts. The characterization results display that the Zn₁Cr₁ catalyst presents more nonstoichiometric Zn–Cr spinel and oxygen vacancies as well as more defects. The Zn₁Cr₁ catalyst also presents better properties of basicity, reducibility and BET surface area which are some other factors for better result of activity evaluation for isobutanol formation by carbon monoxide hydrogenation.

Introduction

Driven by the increasing pressure of energy and environment, more academic studies as well as commercial routes to utilize green sources and chemicals have widely aroused people's extensive interest and attention.^[1–5] The issue of high-efficiency utilizing sustainable resources is one of the century's enormous challenges in the whole world especially in those developing countries where has a huge population density like China and India, etc.^[5,6] Conversion biomass to sustainable substrates offers a promising prospects and readily implemented solution for the demand of high-energy density fuels, fine and specialty chemicals and polymers.^[2,7–10] A convenient route to produce syngas from gasification of biomass (mainly including intractable materials particularly waste). Syngas can also be synthesized from coal and the route is especially suitable for such countries where the resource structure is unbalanced with large stocks of coal but small amount of oil. The major constituents of syngas are CO (carbon monoxide), H₂ (hydrogen), CH₄ (methane) and CO₂ (carbon dioxide).

The conversion from syngas to higher alcohols are identified as a potential process to produce green and high quality transportation fuels. Higher alcohols, typically defined as alcohols possessing two or/and more carbon atoms, attract widespread researcher interest because of their widespread application.^[11] Among higher alcohol, isobutanol is considered one of the most promising additives for gasoline to raise the octane number and the combustion efficiency in automobiles. Besides, isobutanol is widely used as intermediates to synthesize specialty and commodity products in polymer and chemical industries.^[12] Nowadays, isobutanol is chiefly produced by the hydration of the corresponding petroleum-derived alkene over acid catalysts. However, their contribu-

tion to the overall market volume is limited. The demand of isobutanol is also foreseen to dramatically increase in the near future due to the phasing-out of methyl tert-butyl ether (MTBE) and to the raising need for fuels of the growing world population. In order to meet the future demand for isobutanol, a wider application of routes starting from syngas is more appealing since unconventional biomass or coal could be utilized for the production of this feedstock in the future, leading to obvious sustainability advantages.

A wide variety of heterogeneous catalysts have been evaluated in the conversion of syngas into higher alcohol. Generally, catalysts for higher alcohol synthesis can be classified into four categories: Rh-based, Mo-based, modified Fischer-Tropsch catalysts and modified methanol catalyst. The modified methanol catalyst is mostly suitable to produce isobutanol because the alcohol phase mainly contains methanol and isobutanol when syngas react over them.^[35–36] The modified methanol catalyst systems can be broadly classified into two types: low-temperature Cu-based catalysts and high-temperature Cr based catalysts.^[12] The former catalysts are known to be one of the most active systems to form branched alcohols.^[13] However, the average carbon number of alcohol products is lower and the content of ethanol and propanol is

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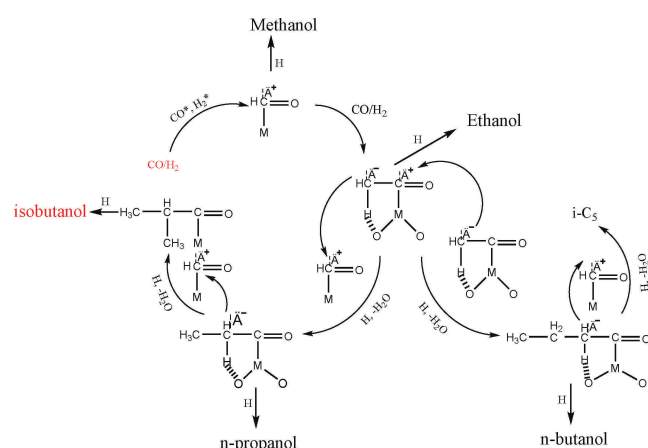
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comparatively high.^[14] The high-temperature catalysts, alkali metal modified Zn–Cr catalyst was another typical catalyst system which was usually used to form isobutanol because of its longer catalyst life, less carbon deposition and concentrated distribution of higher alcohol.^[12,30] They have shown rather high selectivity to isobutanol, over which methanol and isobutanol comprising up to ~95 % in the alcohol phase.^[12,31] The favorable formation of higher alcohol will simplify the approach of isobutanol separation and definitely contribute a lot to the industrialization of isobutanol.

The spinel structure, specifically those containing Zn and Cr ions (Zn–Cr spinel), will be firstly introduced here. These compounds follow the general chemical formula of ZnCr_2O_4 . The Zn^{2+} and Cr^{3+} species locate tetrahedral and octahedral vacancies in the ideal normal spinel, respectively, while the position of Zn^{2+} and half of Cr^{3+} ions is reversed in the ideal inverse spinel.^[12,24] The ideal inverse spinel is thermodynamically less stable than the ideal normal spinel. However, both of ideal normal and inverse spinel is hard to get in reality and the distribution of ions usually lies in between the two extreme structures. Such structure of compound are defined as nonstoichiometric spinels. The ion distribution will dramatically affect the physico-chemical property and of course the activity of catalyst.

The mechanistic basis of isobutanol has been thoroughly studied which is vital to understand the effect of catalyst composition. Some scholars^[13–15] believed that the isobutanol formation mechanism consists of a series of reactions involving. CO is firstly molecularly adsorbed and hydrogenated to form methanol which is not only the initial product but also the reactant to form higher alcohols. C_2 alcohol is formed sluggishly via α -addition reaction (a C_1 intermediate add to the α -C of methanol,) and then β -addition reacts rapidly (the aldol condensation) to produce isobutanol. Chain growth of isobutanol is terminated because of the lack of two β -H required for further β -addition as well as its sterichindrance. The mechanism representing the isobutanol formation was shown in Scheme 1



Scheme 1. Pathway of isobutanol formation

In our previous study, we have revealed the effect of alkali metal promoter for isobutanol formation from carbon monoxide hydrogenation.^[16] We discovered that the presence of potassium could stabilize the oxygen species over the catalyst surfaces and dramatically facilitate the process of isobutanol formation. The differences of catalyst prepared approach were also investigated in our previous experiments.^[17] The catalysts prepared by coprecipitation approach could provide the most uniform chemical and physical properties and the particle size of catalysts was the smallest. We also studied the role of different state ZnO for isobutanol synthesis over Zn–Cr catalyst. We determined the amount of ZnO phase using a linear combination fit in FT-EXAFS figures and the results suggest that the less catalysts own ZnO, the better catalytic performance the catalysts will have.^[34] In industry, the Zn/Cr element ratio for isobutanol formation is 3:1. However, to simplify the research system, we intentionally ignored the effect of excess zinc oxide in our previous study. That means the elemental ratio of Zn/Cr was limited to a certain range: 0.8^[16] or 1.2.^[34] However, the optimal value of Zn/Cr element ratio in a large range (including industrial Zn/Cr molar ratio) for the isobutanol synthesis from syngas has not been studied in our previous experiments. In this paper, we would prepared a series of Zn–Cr catalysts by a facile coprecipitation approach and tested their performances for isobutanol formation by carbon monoxide hydrogenation. These catalysts were then characterized by multiple techniques to explore the essence of different Zn/Cr element ratio for isobutanol synthesis in depth.

Results and Discussion

HRTEM analysis of catalysts

The Morphology and microstructure of catalysts with different Zn/Cr element ratios were investigated by TEM and the images are shown in Figure 1-1 and Figure 1-2. In our previous work,^[17] we have proved that the particle size of zinc oxide catalysts is larger than 20 nm in diameter while the particle size of Zn–Cr spinel catalysts is smaller than 10 nm in diameter. The HRTEM pictures, shown in Figure 1-2, further support this conclusion.

From the TEM images of catalysts, shown in Figure 1-1, it could be found that the particle size of catalysts remained almost the same with increasing Zn/Cr element ratio from 1:3 (Zn_1Cr_3 catalyst) to 1:1 (Zn_1Cr_1 catalyst). Still, the particle size of catalysts became larger when the Zn/Cr element ratio increases to 3:1 (Zn_3Cr_1 catalyst) and reached to the maximum when the Zn/Cr element ratio further increased to 5:1 (Zn_5Cr_1 catalyst).

Depending on the value of interplanar spacing marked in HRTEM images in Figure 1-2, the crystal phase of catalyst could be roughly identified. The results revealed the crystal phase of catalysts changed with different Zn/Cr element ratios. Specifically, the catalyst of Zn/Cr element ratio being 1:3 (Zn_1Cr_3 catalyst) presented both Cr_2O_3 and Zn–Cr spinel

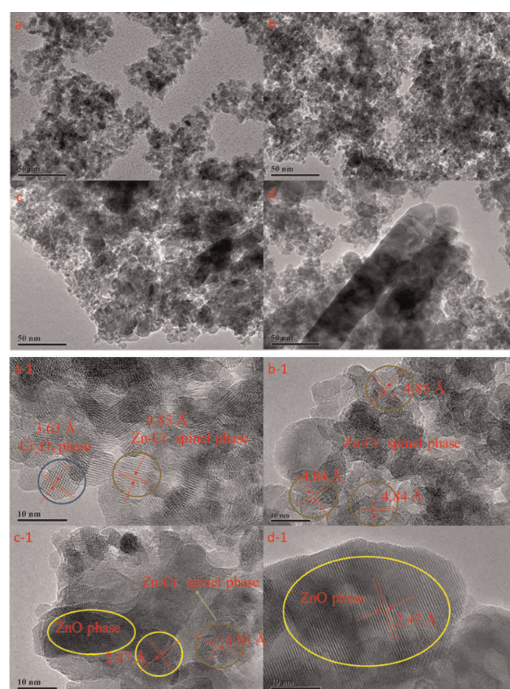


Figure 1. 1) TEM images of the catalysts with different Zn/Cr element ratios, (a) Zn_1Cr_3 , (b) Zn_1Cr_1 , (c) Zn_3Cr_1 , and (d) Zn_5Cr_1 . 2) HRTEM images of the catalysts with different Zn/Cr element ratios, (a-1) Zn_1Cr_3 , (b-1) Zn_1Cr_1 , (c-1) Zn_3Cr_1 , and (d-1) Zn_5Cr_1 .

phase while the Cr_2O_3 phase disappeared and the catalyst just presented Zn–Cr spinel phase with increasing Zn/Cr element ratio to 1:1 (Zn_1Cr_1 catalyst). When the Zn/Cr element ratio further increased to 3:1 (Zn_3Cr_1 catalyst), the ZnO phase with larger particle size appeared and the particle size of ZnO reached to the largest in Zn_5Cr_1 catalyst. The larger particle size of ZnO will of course cover some active sites for isobutanol formation and make negatively effect on the catalyst evaluation results.

X-ray diffractometry of catalysts

The XRD patterns of catalysts with different Zn/Cr element ratios are shown in Figure 2. In compensation, the XRD patterns of ZnO and Cr_2O_3 are also shown in Figure 2. No diffraction peaks for potassium species were found in any of the XRD patterns. No K_2O (JCPDS card 47–1701) or K_2CO_3 (JCPDS card 49–1093) peaks were found, possibly because the small amount of potassium species were highly divided over the catalysts surfaces. The main diffraction peaks are consistent with Cr_2O_3 and Zn–Cr spinel in XRD pattern of Zn_1Cr_3 catalyst.^[18–20] As increasing the Zn/Cr element ratio to 1:1, the Zn_1Cr_1 catalyst just presented nonstoichiometric spinel phase.^[20] The reflections of ZnO phase appeared and became sharper with Zn/Cr element ratio further increasing to 3:1 (Zn_3Cr_1 catalyst) and 5:1 (Zn_5Cr_1 catalyst), suggesting that the particle size of ZnO was becoming larger. That is consistent with the TEM pictures of catalysts.

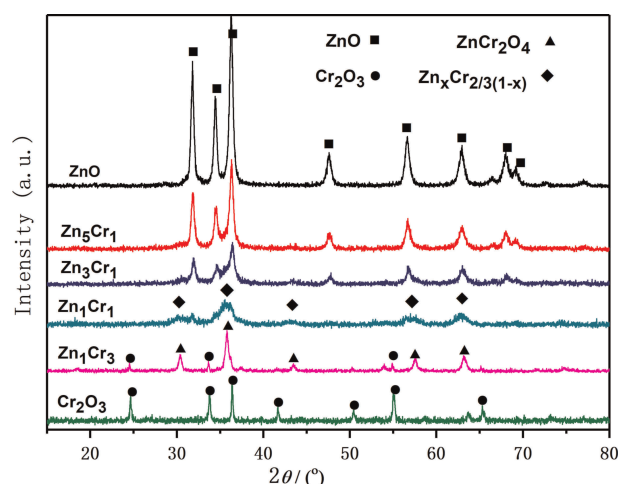


Figure 2. XRD spectra of the catalysts with different Zn/Cr element ratios

In theory, the Zn/Cr element ratio of the ideal normal Zn–Cr spinel is 1:2 (ZnCr_2O_4). It is supposed that the excess Zn species will form ZnO specie. However, it is worth noting that the ZnO characteristic peaks do not exist in the XRD patterns of Zn_1Cr_1 catalysts in which the Zn/Cr element ratio was much more than 1:2. Two reasonable guesses could explain the undetected ZnO phase in XRD patterns, the ZnO phase may be highly dispersed over Zn–Cr spinel surface; or the excess ZnO may have migrated into the spinel structure and make the structure being nonstoichiometric. Tan^[20] and Bertoldi^[21] et al also prepared Zn–Cr samples of Zn/Cr element ratios being 1:1 and only detected the nonstoichiometric Zn–Cr spinel specie in XRD patterns.

XANES of the catalysts

The technique of synchrotron radiation is powerful to detect the situation of ions location in different sites. Recently, the synchrotron radiation technique has been used to investigate the ion distribution in spinel including CuFe_2O_4 , ZnFe_2O_4 , MnFe_2O_4 , etc.^[22–24] As we mentioned above, the metal ions distribution in most spinel structure deviates from the ideal normal/inverse spinel and it is affected by annealing temperature, particle size of samples, and electronic state, etc.^[25] The physico-chemical property of the spinel samples could be tailored by ion distribution in different sites like tetrahedral and octahedral vacancies. Hence, it is crucial to detect the ion distribution in our catalysts.

The normalized XANES spectra of samples are shown in Figure 3. The most striking features due to ion location in the Zn–Cr spinel in XANES spectra appeared around 9680 eV which was named feature A here.^[26] The amplitude of feature A decreased with the Zn/Cr element ratio increasing from 1:3 to 1:1. However, The amplitude of feature A increased when the Zn/Cr element ratio further increasing from 1:1 to 3:1 and 5:1. The amplitude of feature A for ZnO was the strongest while the amplitude of feature A for Zn_1Cr_1 catalyst

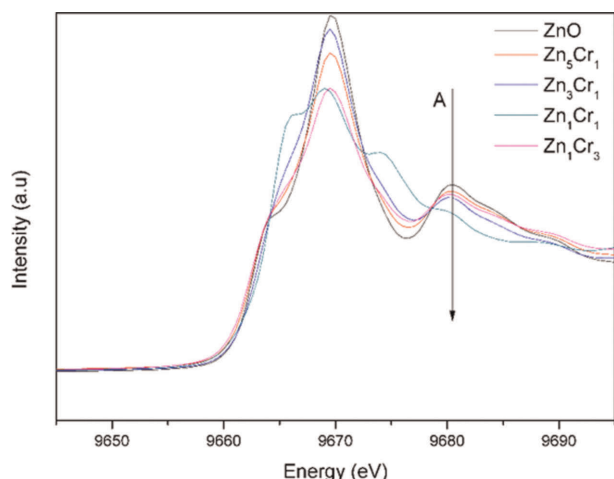


Figure 3. The normalized XANES spectra for the Zn–K-edge of catalysts with different Zn/Cr element ratios.

was the weakest. In a previous study for Zn–Fe spinel, Stewart and coworkers^[26] have confirmed that the feature A decreases in XANES spectrum when the Zn ion migrated from tetrahedral sites (normal spinel) to octahedral sites (nonstoichiometric spinel). It has a highly possibility that the change of feature A has the same reason for Zn–Cr spinel. Some scholars^[24] also got a similar conclusion by processing a simulation Zn ions XANES spectrum for one Zn ion migrated from tetrahedral sites to octahedral sites and the calculation spectrum showed the same change as in Figure 3. Therefore, we can cautiously get a conclusion that the feature A represents the location sites of Zn ions in Zn–Cr spinel, resulting from considerable Zn ions migrating from tetrahedral to octahedral sites. When the amplitude of feature A is weaker, the degree of ion disorder distribution in Zn–Cr spinel will become higher. For our research, the ion distribution in Zn_1Cr_1 catalyst is obviously more disordered and the degree of disorder distribution is followed the order of $\text{Zn}_1\text{Cr}_1 > \text{Zn}_3\text{Cr}_1 > \text{Zn}_1\text{Cr}_3 > \text{Zn}_5\text{Cr}_1 > \text{ZnO}$ catalyst.

EXAFS of the catalysts

The k^3 -weight Zn ion EXAFS spectra and their FT spectra (Fourier transform) for catalysts with different Zn/Cr element ratios are shown in Figure 4 and Figure 5, respectively. From Figure 4, it can be found that the spectra of Zn_5Cr_1 and Zn_3Cr_1 catalysts are similar to ZnO sample while the Zn_1Cr_1 and Zn_1Cr_3 catalysts have another type of Zn–Cr spinel.^[24]

The FT-EXAFS spectra of samples, which could offer the information of atomic distance from Zn atom, are shown in Figure 5. As shown in previous researches,^[27] the distance of first and second shell from Zn atom is 1.97 and 3.45 Å in Zn–Cr spinel in theory. Because of the instrumental error, the first and second peak for FT-EXAFS of Zn ion in Figure 5 appears at around 1.50 and 2.70 Å (not 1.97 and 3.45 Å). We may point out that, the second peak located around 2.70 Å

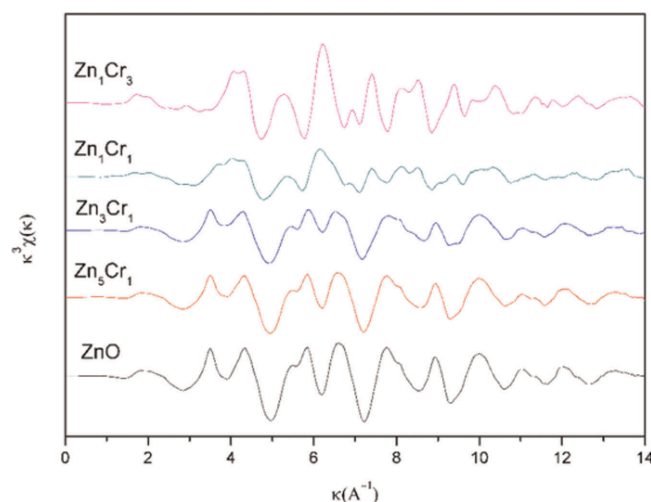


Figure 4. The k^3 -weighted EXAFS signals for the Zn K-edge of catalysts with different Zn/Cr element ratios.

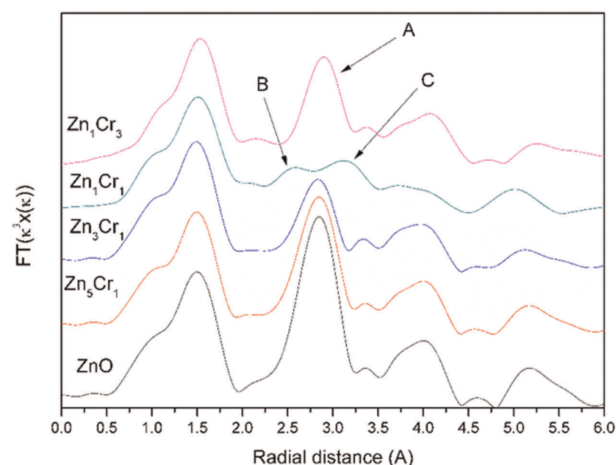


Figure 5. The Fourier transform EXAFS signals for the Zn K-edge of catalysts with different Zn/Cr element ratios.

(feature A) of Zn_1Cr_1 catalyst was split into two peaks, which clearly due to Zn ion disorder distribution in Zn–Cr spinel. Some scholars^[26] have detected a very similar result in Zn ions FT-EXAFS of in Zn–Fe spinel structure. For our samples, when ion disorder distribution appears (nonstoichiometric spinel), which means Zn atoms have migrated into octahedral vacancy, a new shell will definitely appears and results in a new peak (feature B in Figure 5). Not surprisingly, the location of feature B is as same as the second shell distance of Cr ion in Zn–Cr spinel, suggesting the location of Zn and Cr ions have exchanged.^[24] In a word, the appearance of feature B clearly confirms that considerable Zn ions have entered in octahedral vacancies and formed nonstoichiometric spinel in Zn_1Cr_1 catalyst. All EXAFS analysis is consistent with XANES analysis.

XPS characterization of various catalysts

The XPS spectra are shown in Figure 6-1 and Figure 6-2. The location of Zn^{2+} (around 1022.2 eV and 1045.2 eV) and Cr^{3+}

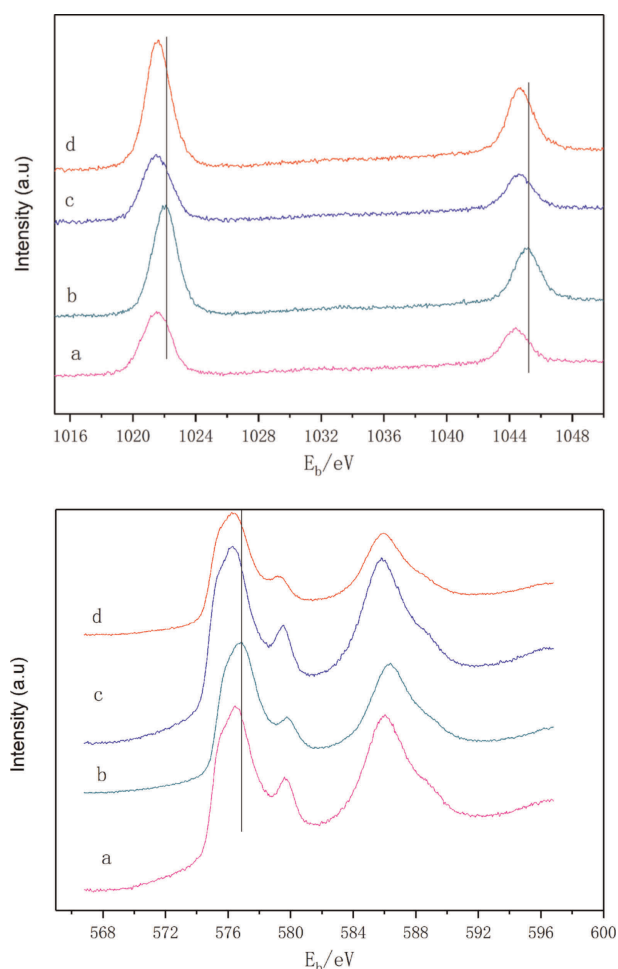


Figure 6. 1) Zn ion XPS analysis of catalysts with different Zn/Cr element ratios, (a) Zn_1Cr_3 , (b) Zn_1Cr_1 , (c) Zn_3Cr_1 , and (d) Zn_5Cr_1 . 2) Cr ion XPS analysis of catalysts with different Zn/Cr element ratios, (a) Zn_1Cr_3 , (b) Zn_1Cr_1 , (c) Zn_3Cr_1 , and (d) Zn_5Cr_1 .

(around 576.3 eV and 586.1 eV) ions implied that Cr and Zn ions over catalysts surface were Zn–Cr spinel or ZnO phase.^[28] All spectra of catalysts also have a Cr^{6+} peak at around 578.2 eV in Figure 6-2, suggesting that catalysts with different Zn/Cr element ratios have been over-oxidized. Take a closely look of Figure 6-1 and Figure 6-2, it can be easily found that the valence of both Zn and Cr ions over the surface of Zn_1Cr_1 catalyst is obviously higher than other catalysts.

Battistoni et al.^[29] found that the binding energy values of Cr and Zn ions for the nonstoichiometric spinel structure is higher than Cr_2O_3 and ZnO species. The change may be result from the different contribution of Madelung potential in spinel and oxide species. The binding energy values of Cr and Zn ions in nonstoichiometric Zn–Cr spinel structure was higher than normal Zn–Cr spinel and ZnO phase, which

resulted from the more defects and oxide vacancies formation in nonstoichiometric spinel structure.^[30] These oxygen vacancies and defects could facilitate the adsorption of the reactants over catalysts surface and obviously promote their catalyst performance.

 H_2 -TPR analysis of the catalysts

The H_2 -TPR curves (temperature program reduction of hydrogen) are shown in Figure 7. The results of ZnO and Cr_2O_3

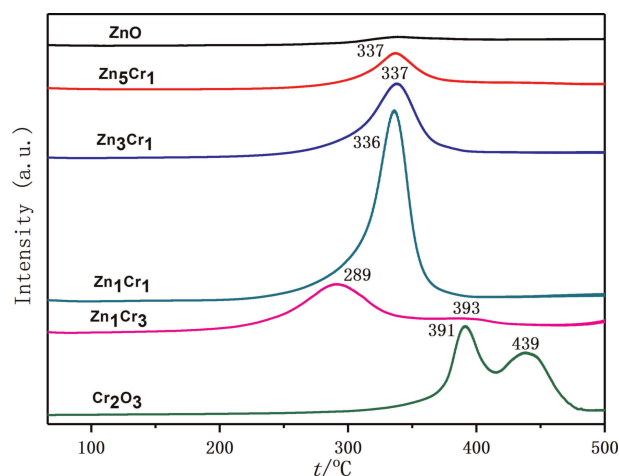


Figure 7. H_2 -TPR curves of the catalysts with different Zn/Cr element ratios.

samples are also present in Figure 7 for comparison. It was hardly to observe the reduction peak over ZnO sample, suggesting that ZnO is very difficult to be reduced in the range of experimental temperature. The two reduction peaks around the temperature of 391 °C and 439 °C over Cr_2O_3 sample was attribute to the reduction of Cr^{6+} to Cr^{3+} and Cr^{3+} to Cr^{2+} , respectively.^[31] Catalysts with different Zn/Cr element ratios had only one reduction peak, resulting from the nonstoichiometric Zn–Cr spinel reduction. Zn^{2+} and Cr^{3+} ions in normal Zn–Cr spinel are highly difficult to be reduced while the reduction of nonstoichiometric spinel catalysts are much easier because of the appearance of defects and oxygen vacancies.^[20] The value of hydrogen consumption was shown in Table 1. It

Table 1. Texture parameters of catalysts with different Zn/Cr element ratios

Catalysts	$S_{\text{BET}}(\text{m}^2\text{g}^{-1})$	$V_p(\text{cm}^3\text{g}^{-1})$	$R_p(\text{nm})$	$C_{\text{H}_2}(\text{mmol/g})$
Zn_1Cr_3	96.1	0.401	13.7	0.62
Zn_1Cr_1	101.2	0.536	14.4	1.53
Zn_3Cr_1	65.4	0.315	10.2	0.53
Zn_5Cr_1	45.5	0.242	9.01	0.21

can be found that hydrogen consumption for the Zn_1Cr_3 , Zn_1Cr_1 , Zn_3Cr_1 and Zn_5Cr_1 catalysts was ~0.62 mmol/g, ~1.53 mmol/g, ~0.53 mmol/g, and ~0.21 mmol/g, respectively. It

Table 2. Catalyst performance of the catalysts with different Zn/Cr element ratios, uncertainties in the least significant digit are given in parentheses

Catalysts	CO Conversion (%)	STY (g/ml h) ^a	Alcohols distribution (wt %)				
			MeOH	EtOH	n-PrOH	i-BuOH	C ₅₊ alcohol
Zn ₃ Cr ₁	19.(3)	0.06(2)	82.(6)	1.6(7)	3.0(7)	11.(7)	0.9(6)
Zn ₃ Cr ₁	29.(6)	0.07(4)	77.(3)	1.8(1)	3.0(1)	16.(5)	1.3(8)
Zn ₁ Cr ₁	32.(6)	0.08(4)	70.(3)	1.4(7)	2.9(1)	24.(1)	1.2(2)
Zn ₁ Cr ₃	11.(3)	0.01(0)	81.(5)	0.9(1)	2.3(4)	14.(2)	1.0(5)
ZnO	15.(5)	0.02(3)	76.(6)	11.(3)	8.2(9)	3.1(3)	0.6(9)
Cr ₂ O ₃	0	0	0	0	0	0	0

Reaction conditions: Temperature = 400 °C, Pressure = 10 MPa, GHSV = 3000 h⁻¹; a: the TOS (time-on-stream) for calculating the STY (space time yield) of alcohols was 8 h.

further confirmed that the amount of nonstoichiometric spinel came to the maximum in Zn₁Cr₁ catalyst.

Texture parameters of catalysts

The information of S_{BET} surface areas (Brunauer–Emmett–Teller) and V_p (pore volumes) of different catalysts are shown in Table 1. The S_{BET} of catalysts increased from 96.1 m²g⁻¹ to 101.2 m²g⁻¹ with increasing the Zn/Cr element ratio from 1:3 to 1:1, while the S_{BET} further decreased from 101.2 m²g⁻¹ to 65.4 m²g⁻¹ and 45.5 m²g⁻¹ with increasing the Zn/Cr element ratio from 1:1 to 3:1 and 5:1, respectively.

The S_{BET} obviously increased by increasing Zn/Cr element ratio from 1:3 to 1:1. It is most likely result from the appearance of nonstoichiometric phase. In a previous study, Bertoldi et al.^[21] discovered the S_{BET} of catalysts would increase obviously when Zn ions inside the normal Zn–Cr spinel, while the appearance of ZnO in catalysts would decrease the S_{BET} of catalysts. In our study, the S_{BET} of catalysts also dramatically decreased when ZnO phase appeared in Zn₃Cr₁ and Zn₅Cr₁ catalysts. That may be result

from the larger particle size of ZnO phase which have been confirmed by TEM and XRD results.

The information of V_p (pore volumes) and R_p (pore radii) is also shown in Table 1. It can be found easily that both the V_p and R_p of Zn₁Cr₁ catalyst are bigger than the others. The bigger V_p and R_p of Zn₁Cr₁ is more beneficial for isobutanol diffusion from inner pore of catalyst, which of course benefit a lot for isobutanol formation in Zn₁Cr₁ catalyst.

CO₂-TPD analysis of catalysts

The alkaline property of different catalysts are estimated by CO₂-TPD and the results are showed in Figure 8. Each CO₂-TPD curve has a weak and a strong adsorption of CO₂, meaning all of catalysts contains weak and strong basic sites. It is also evident that the amount of CO₂ adsorption increased with increasing Zn/Cr element ratio from 1:3 to 1:1 and decreased with further increasing Zn/Cr element ratio to 3:1 and 5:1. Theoretically, the alkaline property is closely connected with the catalyst surface chemistry, the enhanced basic property should result from the surface defects and oxygen vacancies in nonstoichiometric spinel.

The mechanism of isobutanol formation consists of a series of reactions involving, containing continuous reactions of α -addition and β -addition. β -addition is a processing of aldo-condensation and it is strongly influenced by alkaline property of catalysts surface. Zn₁Cr₁ catalyst could offer more amount of alkaline sites and that favours the formation of isobutanol.

Catalytic activity measurement

The evaluation results of different catalysts for isobutanol synthesis are shown in Table 2. The results of catalytic activity measurement of ZnO and Cr₂O₃ samples are also shown in Table 2 as reference. The carbon balance was all ranged from ~96 % to 105 % in our experiments, suggesting the reliability of our experiments.

In Table 2, none of any catalytic activity was observed over Cr₂O₃ sample. ~15.5 % CO was converted over ZnO sample and the selectivity of isobutanol contributed only 3.13 % in alcohols phase. For the different catalysts, the CO

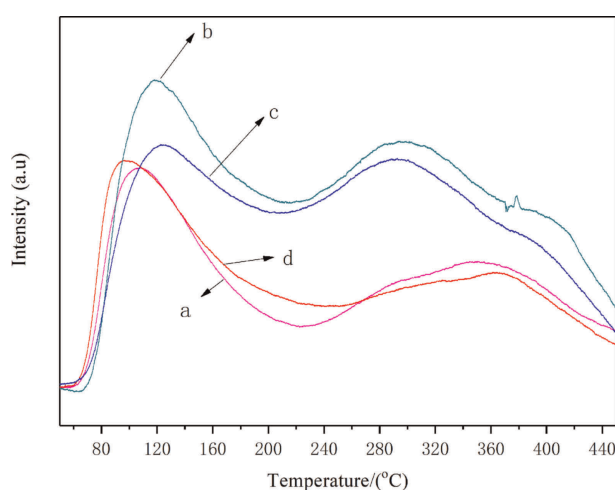


Figure 8. The CO₂-TPD curves of the catalysts with different Zn/Cr element ratios, (a) Zn₁Cr₃, (b) Zn₁Cr₁, (c) Zn₃Cr₁, and (d) Zn₅Cr₁.

conversion over Zn_1Cr_3 catalyst was 11.3 % and the selectivity of isobutanol in alcohol phase was 14.2 %; the catalyst with Zn/Cr element ratio being 1:1 (Zn_1Cr_1 catalyst) presented an excellent catalyst performance over which the CO conversion was enhanced to 32.6 % and the selectivity of isobutanol was greatly improved to 24.1 %; when the Zn/Cr element ratio increased from 1:1 to 3:1, the CO conversion over Zn_3Cr_1 catalyst decreased a little to 29.6 % and the selectivity of isobutanol also lowered to 16.5 %; when the Zn/Cr element ratio further increased to 5:1, the CO conversion and isobutanol selectivity over Zn_5Cr_1 catalyst further decreased to 19.3 % and 11.7 %, respectively.

It can be concluded that the Zn/Cr element ratio significantly affects the activity and selectivity for isobutanol formation from carbon monoxide hydrogenation reaction. The Zn_1Cr_1 catalyst shows the best catalyst performance including CO conversion and isobutanol selectivity in which the nonstoichiometric spinel becomes the main phase. Some scholars believed that the ZnO phase was the active sites to form methanol^[32,33] and the nonstoichiometric spinel was responsible for β -addition to form isobutanol.^[21,30,31] The mechanism of isobutanol formation consists of a series of reactions involving, the original product is methanol which is also the reactant to produce isobutanol. It is worth noting that, the process of methanol formation is a rapid reaction on Zn–Cr catalysts, while the process of isobutanol formation from methanol is more difficult and more essential for our experiments. That means the latter active sites (nonstoichiometric spinel) is more significant than the former active sites (ZnO species). Furthermore, comparing to ZnO and normal Zn–Cr spinel, the nonstoichiometric Zn–Cr spinel catalysts is a metastable structure which presents more oxygen vacancies and defects and these vacancies and defects should obviously facilitate the adsorption of reactant molecule (CO and H_2) over the catalyst surface and benefit the catalyst performance a lot. The Zn_1Cr_1 catalyst also presents some other better properties for isobutanol formation over Zn–Cr catalysts from carbon monoxide hydrogenation such as better reducibility, larger S_{BET} surface area and better alkaline.

Conclusions

In the work, we prepared a few catalysts of different Zn/Cr element ratios by a coprecipitation approach and test their catalyst activity for isobutanol formation from carbon monoxide hydrogenation. Among these different catalysts, The Zn_1Cr_1 catalyst of Zn/Cr element ratio being 1:1 presented outstanding activity for iso-butanol formation from the carbon monoxide hydrogenation, carbon monoxide conversion and isobutanol selectivity reached to the maximum to ~32 % and ~24 %, respectively.

A proper Zn/Cr element ratio over Zn–Cr catalysts could greatly improve their catalyst performance. The Zn_1Cr_1 catalyst presents the most serious disorder distribution in nonstoichiometric Zn–Cr spinel. The nonstoichiometric Zn–Cr spinel catalysts is a meta-stable structure which

presents more oxygen vacancies and defects. These vacancies and defects should obviously facilitate the adsorption of reactant molecule (CO and H_2) over the catalyst surface and benefit the catalyst performance a lot. The Zn_1Cr_1 catalyst also presents some other better properties for isobutanol formation over Zn–Cr catalysts from carbon monoxide hydrogenation such as better reducibility, larger S_{BET} surface area and better alkaline.

Experimental Section

Catalysts preparation

The catalysts of different Zn/Cr element ratio were prepared by a coprecipitation approach. The mixture solution of Zn ($(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$) and Cr($(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$) was named solution A while the solution of $(\text{NH}_4)_2\text{CO}_3 \cdot \text{CH}_3\text{NO}_2$ was named solution B. The solution A and solution B were put together into a violently stirred container at the temperature of 60 °C and the pH of 11. The precipitate was settled at room temperature for about 12 hours and washed thoroughly with deionized water three times to make sure the pH of solution was 7. The precursor was calcined at 350 °C for 6 hours after drying at 120 °C for 12 hours. The catalyst was powdered thoroughly and impregnated 3.0 wt% potassium metal using K_2CO_3 as a precursor, then it was dried at 120 °C for 12 hours and annealing at 400 °C for 6 hours. The sample was finally broken into 30–40 mesh. The different Zn/Cr element ratios of catalysts were controlled by regulating the amount of raw material. In this paper, catalysts were denoted as “ Zn_iCr_j ” for the catalyst with Zn/Cr element ratio being 1:1. The other catalysts were similar.

Characterization

The picture of HRTEM (high resolution transmission electron microscopy) was investigated by a field emission transmission electron microscope (JEM-2100F) at 200 kV with a resolution of ~0.2 nm.

The XRD patterns (X-ray powder diffraction) were measured by the Bruker D8 powder diffractometer using Cu K (40 kV, 100 mA) radiation and Ni mono-chromator. The 2θ range was from 10° to 80°.

The XPS spectrum (X-ray photoelectron spectroscopy) were conducted using an AXIS ULTRA DLD instrument. The Al K_{α} monochromatized line (1486.6 eV) was used in a vacuum situation (the pressure being 2×10^{-8} Pa).

The nitrogen adsorption and desorption isotherms determined using a constant-volume adsorption apparatus (Micromeritics, ASAP-2400). The pore size and distribution was evaluated by BJH formula (Barrett–Joyner–Halenda) while the S_{BET} (specific surface area) was calculated using the BET method.

The H_2 -TPR (Temperature Programmed Reduction of Hydrogen) was processed on a chemisorption instrument (TP-5050) in the range from 50 °C to 550 °C. The 100 mg catalyst was heated at 300 °C in Argon flow for eliminating physical adsorbed impurities and then cool down to 50 °C. The Argon flow was replaced by the gas of 10 % H_2/Ar mixture. Then, the sample was heated at the rate of 5 °C/min to 550 °C. The vent gas was analyzed by the thermal conductivity detector. The detailed quantification of H_2 consumption in the H_2 -TPR experiments presented in the supporting information section.

The CO₂-TPD (The Temperature Programmed Desorption of carbon dioxide) was also processed on the chemisorption instrument (TP-5050) in the range from 50 °C to 550 °C. The zeolite and activated carbon were firstly used to pure the gas. The sample was reduced by reducing gas (10 % H₂ in Ar) at 400 °C for 3 hours. Then, the sample was saturated by CO₂ flow at 50 °C. Then, the sample was heated at the rate of 5 °C/min to 550 °C under Argon flow. The output gas was investigated by the thermal conductivity measurements.

The XAS (X-ray Absorption spectrum) were processed at the BL14 W1 beamline in Shanghai Synchrotron Radiation Facility. The Si (111) was equipped as monochromator at BL14 W1 beamline. The spectra were recorded in the transmission mode at the zinc and chrome K-edges. The experiments were performed with an electron energy of 3.5 GeV with a 200 mA constant current.

The XAS data was analyzed using the software of ATHENA. For all of curves, the pre-edge and post-edge background were firstly removed and then normalized to 1. The FT-spectra (Fourier transform) was gotten in the Hanning window in the range of 3.5–13.5 Å⁻¹ for Zn atom and 3.5–12.5 Å⁻¹ for Cr atom. The instrumental error make the first and second peak for FT-EXAFS of Zn ion in Figure 5 move left ~0.5 Å which didn't affect analysis

Catalytic isobutanol evolution

The catalyst performance was evaluated in a fixed-bed reactor using 5 mL catalyst. The catalyst was firstly reduced at 360 °C by reduction gas (H₂/N₂ = 10/90) for 4 hours. The molecular ratio of H₂/CO in syngas is 2.6 approximately which mainly includes CO, H₂, CO₂ and CH₄, etc. The reactions were implemented at 400 °C, 3000 h⁻¹ and 10 MPa. The liquid-production was analysed by the GC 4000 gas chromatograph. The gas-production and gas-reaction were analysed by the thermal conductivity detector (TCD). The CH_x mixtures including C₁ to C₅ alkane were analysed by a flame ionization detector. The alcohol production was analysed by the GC-7 A which equipped a flame ionization detector. The computational formulas of CO conversion, alcohol yield and alcohol selectivity was shown in supporting information section.

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